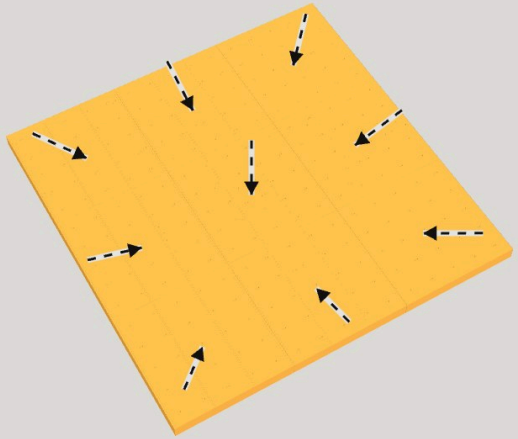
1





1× FRM-01a



1× FRM-01b



1× FRM-01c



1× FRM-01d



1× FRM-01e



1× FRM-01f



1× FRM-01g



1× FRM-01h



1× FRM-01i



M2 × 10

Bolt the deck together

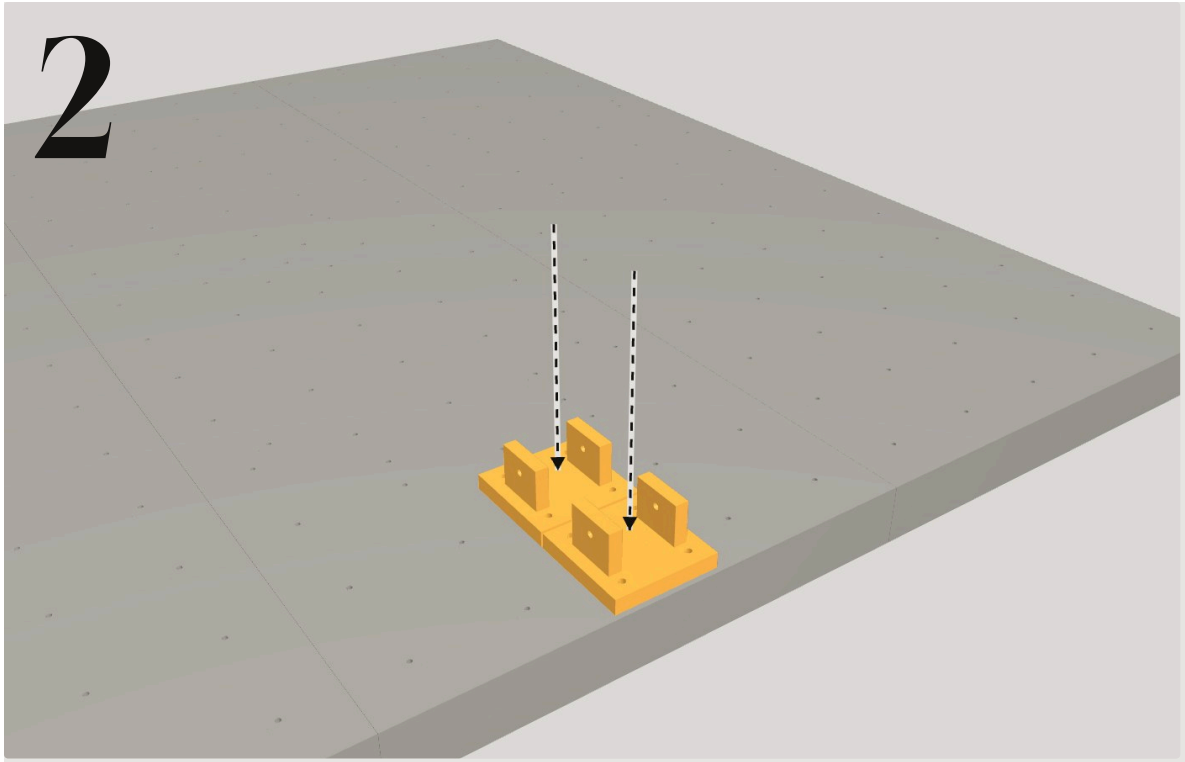
Stage 03 · Frame · step 1 of 24

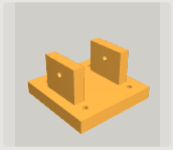
- 1 × Deck tile SW (x -300..-100, y -300..-100) FRM-01a · printed · STL
- 1 × Deck tile S (x -100..100, y -300..-100) FRM-01b · printed · STL
- 1 × Deck tile SE: power plate (x 100..300, y -300..-100) FRM-01c · printed · STL
- 1 × Deck tile W (x -300..-100, y -100..100) FRM-01d · printed · STL
- 1 × Deck tile C: table post + motor (x -100..100, y -100..100) FRM-01e · printed · STL
- 1 × Deck tile E: M4 motor + saddles (x 100..300, y -100..100) FRM-01f · printed · STL
- 1 × Deck tile NW: controller tray (x -300..-100, y 100..300) FRM-01g · printed · STL
- 1 × Deck tile N (x -100..100, y 100..300) FRM-01h · printed · STL
- 1 × Deck tile NE (x 100..300, y 100..300) FRM-01i · printed · STL

1. Tiles upside down (top face on the table, ribs up), in their places as seen from below: east and west swap, so E tiles lie on your left when N points away.
2. Butt neighbouring walls; M2 × 10 through each pair of seam holes (3 per edge), washer and nut on the other side. Finger-tight everywhere first, then tighten from the centre tile outward.
3. Turn the deck over, onto the table where it will live (+X, east, toward where the camera goes; leave 450 mm free beyond that edge for the outrigger).

Checklist step M11.3 · Bolt the deck together (stage 03 · Frame)

2





2× FRM-09d



4× M2 × 12

Outrigger saddles

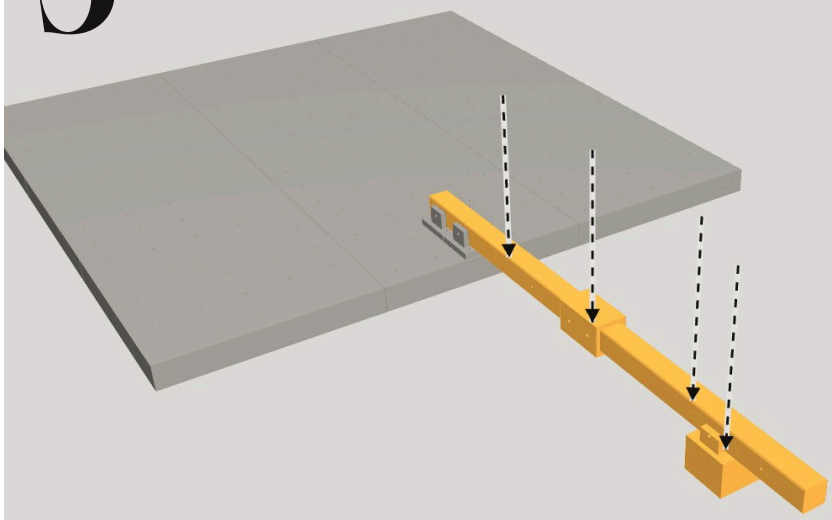
Stage 03 · Frame · step 2 of 24

2 × Outrigger saddle FRM-09d · printed · STL

1. Saddles on their pilots in the east tile (x = 237 and 278, on the +X line): 4 × M2 × 12 each.

Checklist step M12.1 · Outrigger, camera pillar and scan post (stage 03 · Frame)

3





1× FRM-02a



1× FRM-02b



1× FRM-09i



1× FRM-09f



M2 × 16

Outrigger and end foot

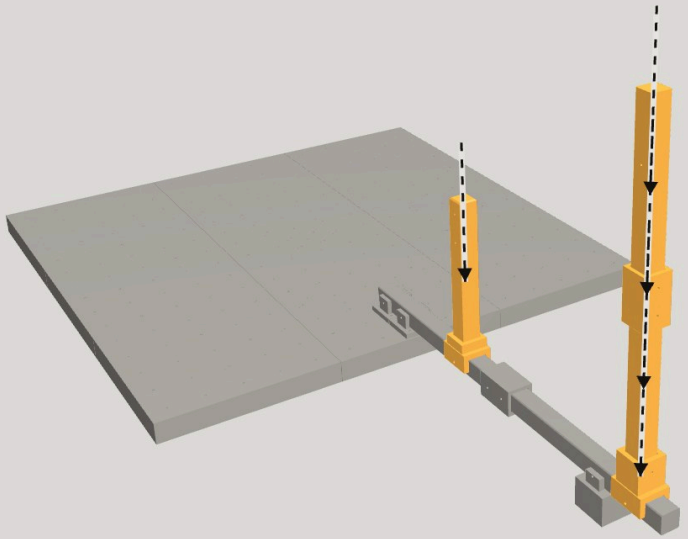
Stage 03 · Frame · step 3 of 24

- 1 × Outrigger bar 1, 250 mm (x 215..465) FRM-02a · printed · STL
- 1 × Outrigger bar 2, 250 mm (x 465..715) FRM-02b · printed · STL
- 1 × Bar coupler: outrigger, scan post FRM-09i · printed · STL
- 1 × Outrigger end foot FRM-09f · printed · STL

1. Outrigger: bars 1 and 2 butted inside a coupler, M2 × 16 through the coupler into each bar's pilots. Lay it in the saddles, inner end at x = 215, the far end on the end foot (standing on the table at x = 628). Every clamp's hole now lines up with a pilot in the bar: M2 × 16 through each saddle cheek and the end foot.

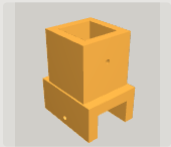
Checklist step M12.1 · Outrigger, camera pillar and scan post (stage 03 · Frame)

4





1× FRM-09h



1× FRM-09e



1× FRM-02c



1× FRM-09i



1× FRM-02d



M2 × 16

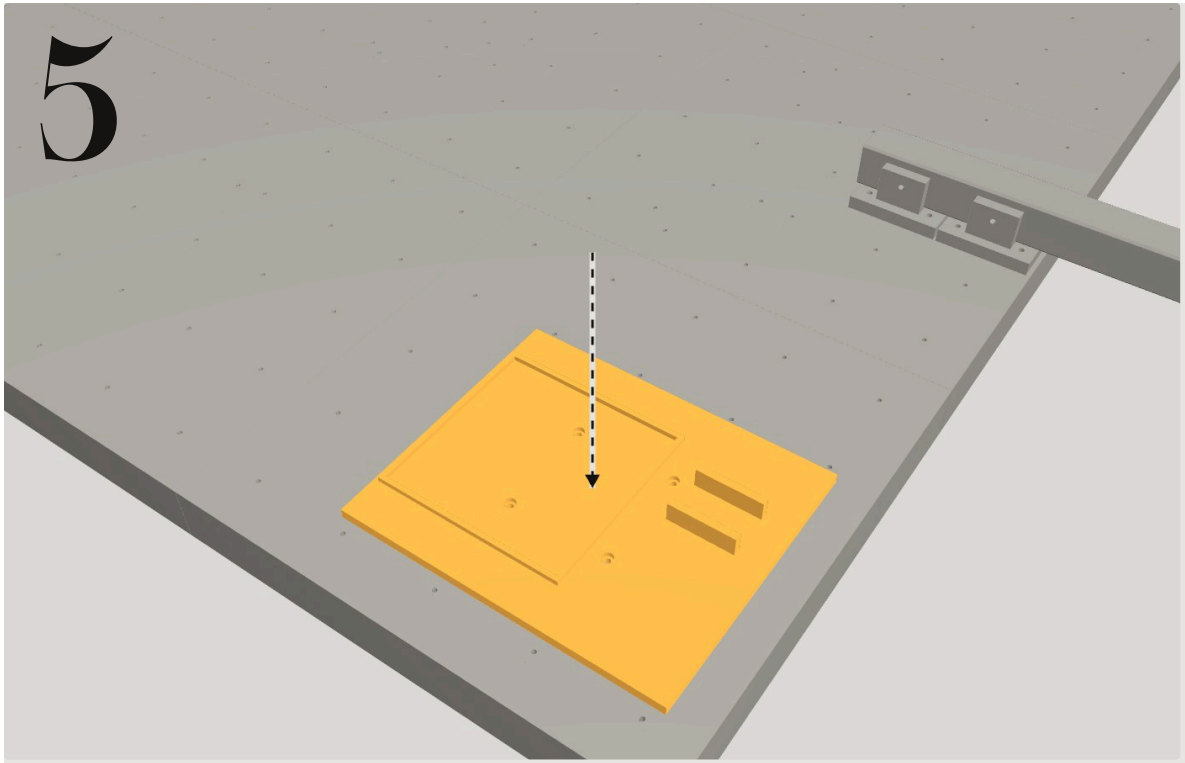
Camera pillar and scan post

Stage 03 · Frame · step 4 of 24

- 1 × Measuring-camera pillar on the outrigger FRM-09h · printed · STL
- 1 × Scan-post socket on the outrigger FRM-09e · printed · STL
- 1 × Scan post bar 1, 175 mm (z 36..211) FRM-02c · printed · STL
- 1 × Bar coupler: outrigger, scan post FRM-09i · printed · STL
- 1 × Scan post bar 2, 175 mm (z 211..386) FRM-02d · printed · STL

1. Camera pillar (FRM-09h) over the outrigger at its pilot (x = 391.8); post socket over the pilot at x = 674.4, M2 × 16 each. Scan post: bar 1 into the socket (M2 × 16), bar 2 on top through the second coupler. No tall post at the measuring camera: it would stand in the scan camera's view.

Checklist step M12.1 · Outrigger, camera pillar and scan post (stage 03 · Frame)





1× PWR-18



4× M2 × 12

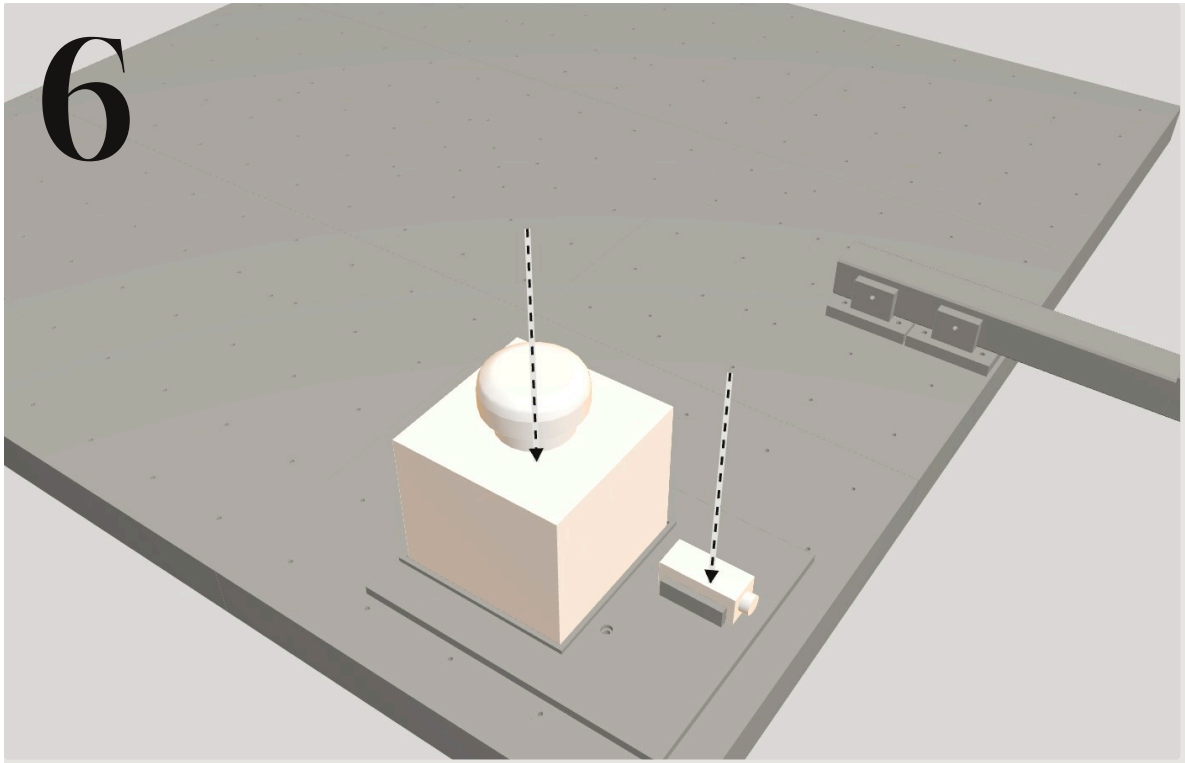
Power plate

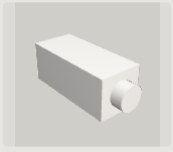
Stage 03 · Frame · step 5 of 24

1 × Power plate (jack + e-stop box) PWR-18 · printed · STL

1. Power plate: 4 × M2 × 12 through its counterbored holes into the SE tile's pilots (centre 220, −220), jack socket toward the +X edge. Lift the e-stop box off its tape to reach the two holes under it, then press it back.

Checklist step M12.2 · Mount the plate and tray (stage 03 · Frame)





1× PWR-30



1× PWR-28

Jack and e-stop box

Stage 01 · Power · step 6 of 24

Only 24 V inside the machine: the wall adapter stays outside it. Before touching any screw terminal: press the e-stop and pull the adapter out of the wall. Wire everything with the adapter unplugged; it goes into the wall last.

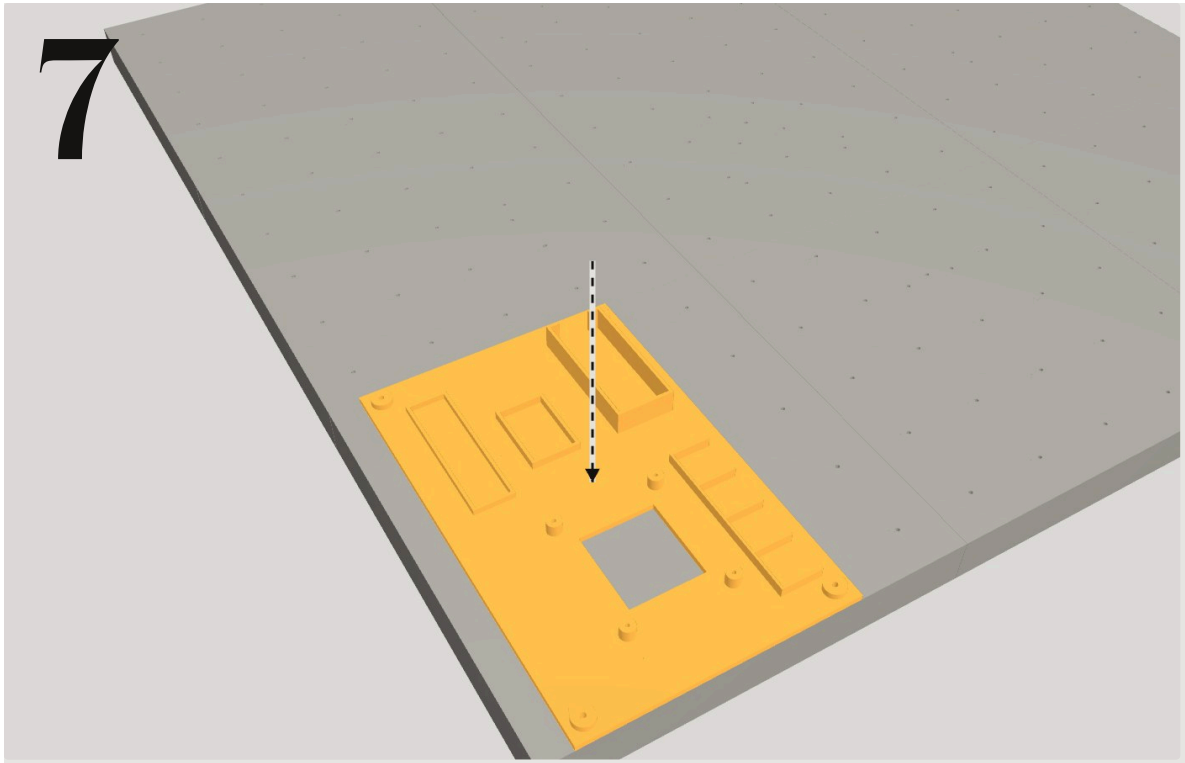
- 1 × 24 V 3 A wall adapter with screw-terminal DC jack PWR-30 · bought
- 1 × Emergency stop in its own box, latching, 1 NO + 1 NC, screw terminals PWR-28 · bought

1. Jack into its frame on the plate, round socket toward the +X edge, screws toward the e-stop.
2. E-stop box into its rim on double-sided tape, button up. Lid off for wiring.

Done on the bench in stage 01, before the plate goes on the deck; shown here in place. Wiring is in the wiring guide.

Checklist step M05.1 · Plate hardware (stage 01 · Power)

7



Controller tray

Stage 03 · Frame · step 7 of 24

1 × Controller tray (SKR Pico + ESP32-S3 + opto + LED driver) CTL-10b · printed · STL

1. Controller tray: 4 × M2 × 12 through its corner bosses into the NW tile's pilots (centre −200, 220).

Checklist step M12.2 · Mount the plate and tray (stage 03 · Frame)

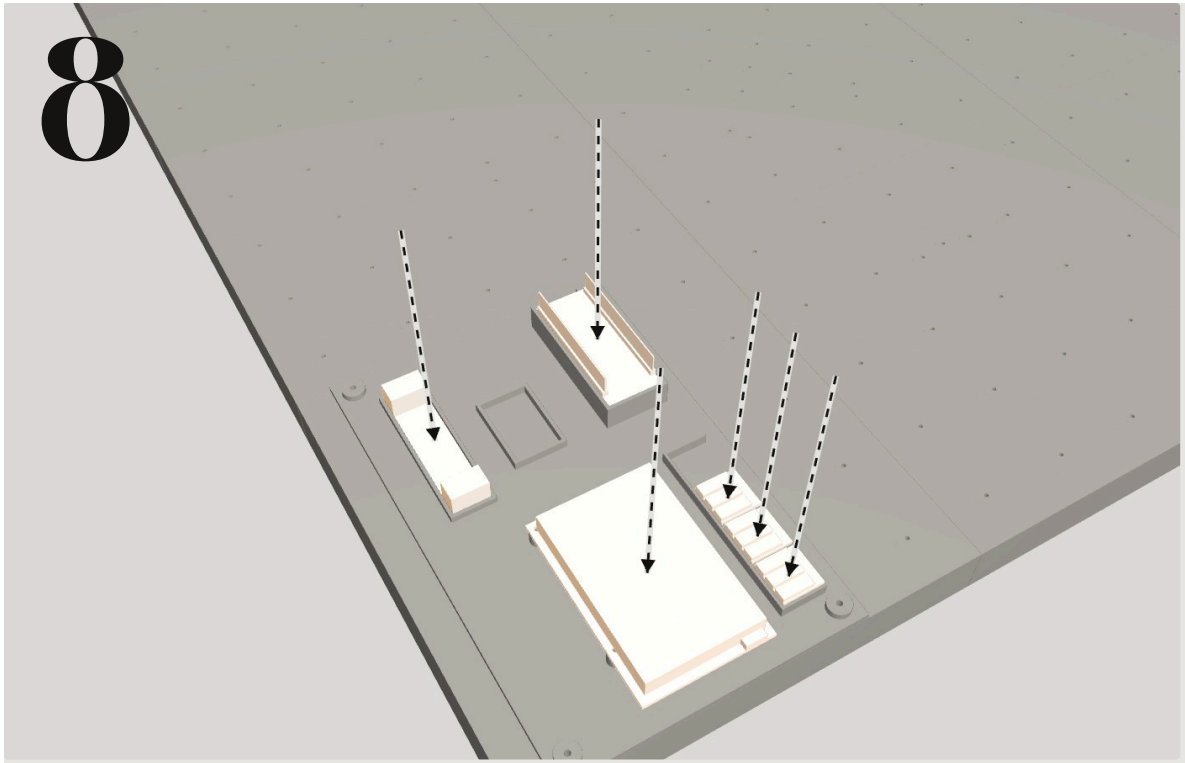


1× CTL-10b



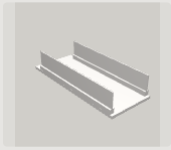
4× M2 × 12

8





1× CTL-01



1× CTL-02



1× PWR-10



3× PWR-20



M2 × 8

Boards on the tray

Stage 02 · Controller and buttons · step 8 of 24

Never plug or unplug motors with 24 V on.

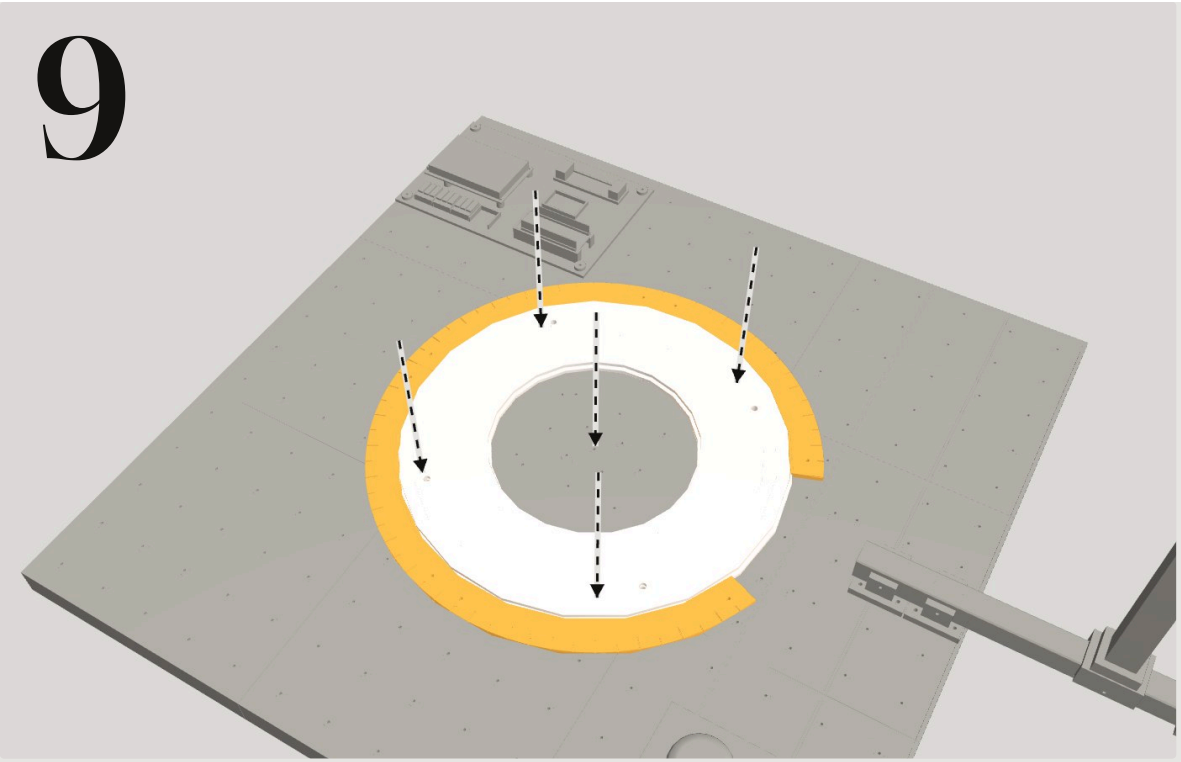
- **1 ×** BIGTREETECH SKR Pico V1.0 (RP2040, 4 × TMC2209 UART, USB-C) CTL-01 · bought
- **1 ×** ESP32-S3-DevKitC-1 (owned) CTL-02 · bought
- **1 ×** Optocoupler module PC817, 24 V input, screw terminals PWR-10 · bought
- **3 ×** Lever wire connectors, WAGO 221 assortment PWR-20 · bought

1. Drive M2 × 8 into the 4 SKR Pico standoffs once; mount the board (holes 58 × 49 mm), USB-C toward the tray edge, a washer under each head.
2. ESP32-S3 (flashed in M09.1): a square of foam tape on the pedestal, then the board upside down (module can on the tape, header pins up), USB end at the opening.
3. Opto module (PWR-10) into the long frame, double-sided tape under it, terminals up. The short frame is for the LED driver (M18.1). Keypads: peel and stick into the panel recesses, tails through the slots.
4. Lever connectors: MOT+ and MOT− (from M05.2) and a third 221-413 for the keypad GND (M09) slide into the pockets along the tray's south edge, wire openings out, levers up. The fourth pocket is spare.

Done in stage 02, before the tray goes on the deck; shown here in place.

Checklist step M10.1 · Print and mount (stage 02 · Controller and buttons)

9





1× LAR-01



4× FRM-09c



2× M2 × 10

Lazy Susan and ring scale

Stage 04 · Light arm (M4) · step 9 of 24

Before touching a motor plug or a screw terminal: press the e-stop and unplug the adapter.

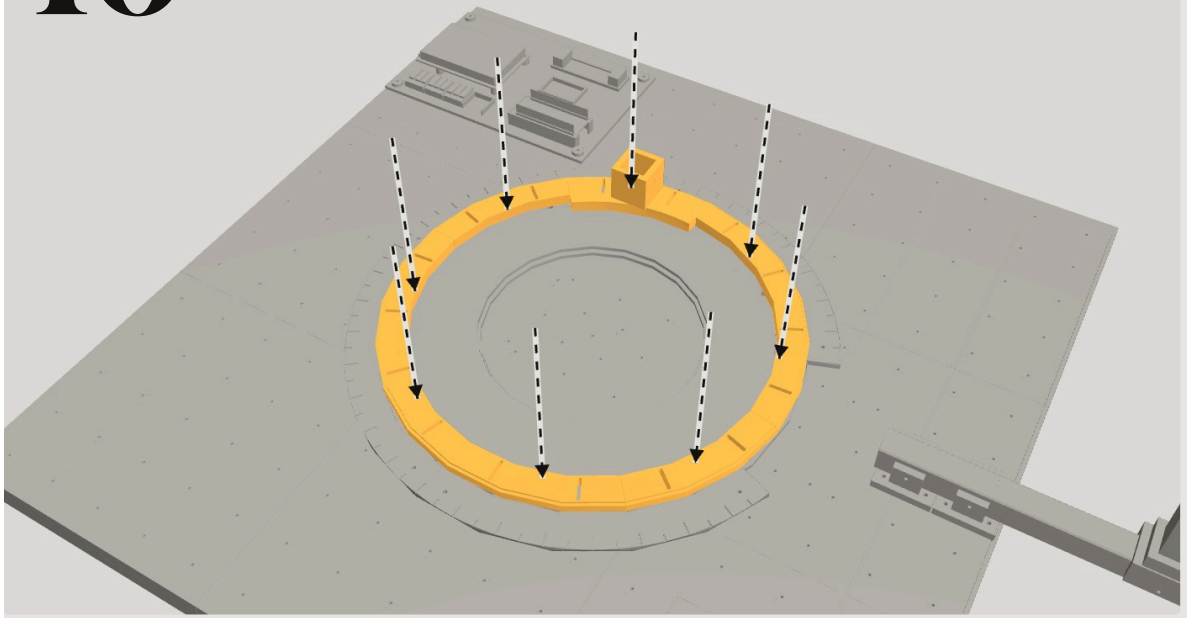
1 × 12 in lazy Susan bearing LAR-01 · bought

4 × Ring scale + lazy-Susan clamp, 80 deg FRM-09c · printed · STL

1. Lazy Susan on the deck centre, its bottom plate inside the ring of 8 pilot holes (r 163). Ring-scale segments end to end from 20° to 340° (0° = +X, toward the camera; the gap is the M4 motor): each inner wall against the bottom plate's edge, its lip over the plate; 2 × M2 × 10 + washer into the deck pilots. The lazy Susan needs no screws of its own.

Checklist step M13.3 · Lazy Susan, rim and belt (stage 04 · Light arm (M4))

10





7× LAR-08a



1× LAR-08c



M2 × 12

Rim and arm segment

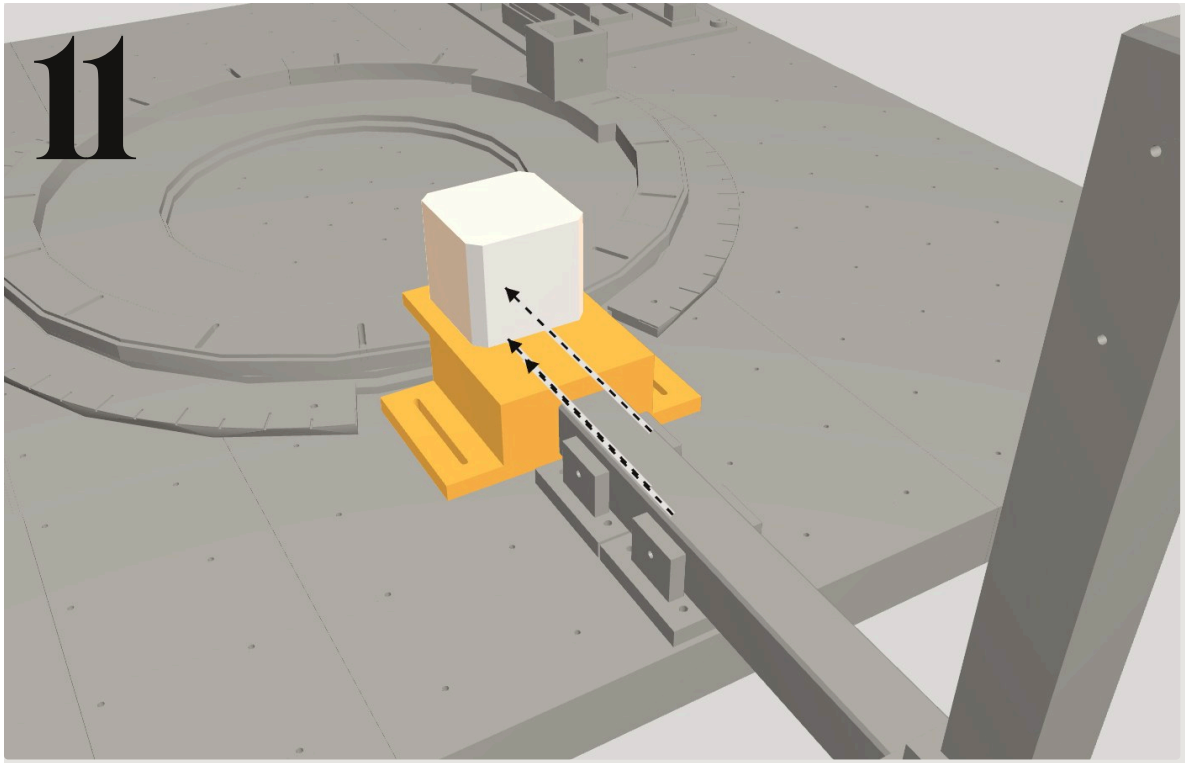
Stage 04 · Light arm (M4) · step 10 of 24

7 × Belt rim segment LAR-08a · printed · STL

1 × Arm segment: rim segment + upright socket LAR-08c · printed · STL

1. Rim segments + the arm segment on the top plate, closed ring: turn the ring so an arm-segment slot lands on a plate hole; M2 × 12 + wide printed washer + nut through every slot that lands on a hole; CA-glue the rest (the arm segment's whole underside).
2. CA-glue the GT2 belt teeth-out on the 6 mm band, its edges flush with the band's edges, ends butted, no gap; it ties the segments together.

Checklist step M13.3 · Lazy Susan, rim and belt (stage 04 · Light arm (M4))




1× LAR-08b


1× LAR-02


1× LAR-04


M2 × 12


4× M3 × 8

Ring motor

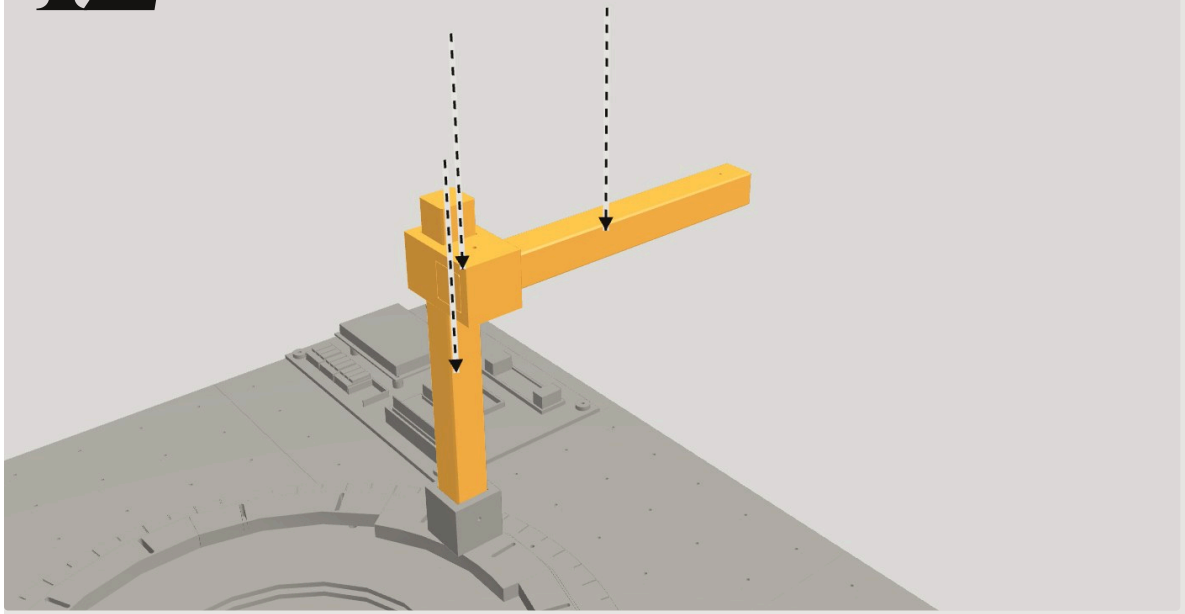
Stage 04 · Light arm (M4) · step 11 of 24

- 1 × M4 motor mount at 0 deg, pinion meshes the rim LAR-08b · printed · STL
- 1 × NEMA 17 stepper 42 Ncm 1.5 A, 38 mm LAR-02 · bought
- 1 × GT2 20T pulley 5 mm bore LAR-04 · bought

1. E-stop pressed. This is the bench motor from M08.3: unplug it from the SKR Pico X socket only now, mount it, then plug it back into X with its cable along the deck edge.
2. Motor on the mount plate (shaft down through the slot, 4 × M3 × 8); 20T pulley on the shaft, grub on the D flat, teeth level with the rim belt (its flanges pass above and below the band).
3. Mount at 0° (+X), base outside the lazy Susan, its two slots over the two deck pilots; slide it in until the pulley meshes the rim belt fully, then M2 × 12 + washer through each slot.

Checklist step M13.4 · Motor (stage 04 · Light arm (M4))

12

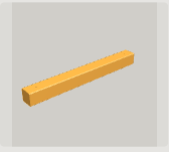




1× LAR-08f



1× LAR-08e



1× LAR-08g



M2 × 16

Arm

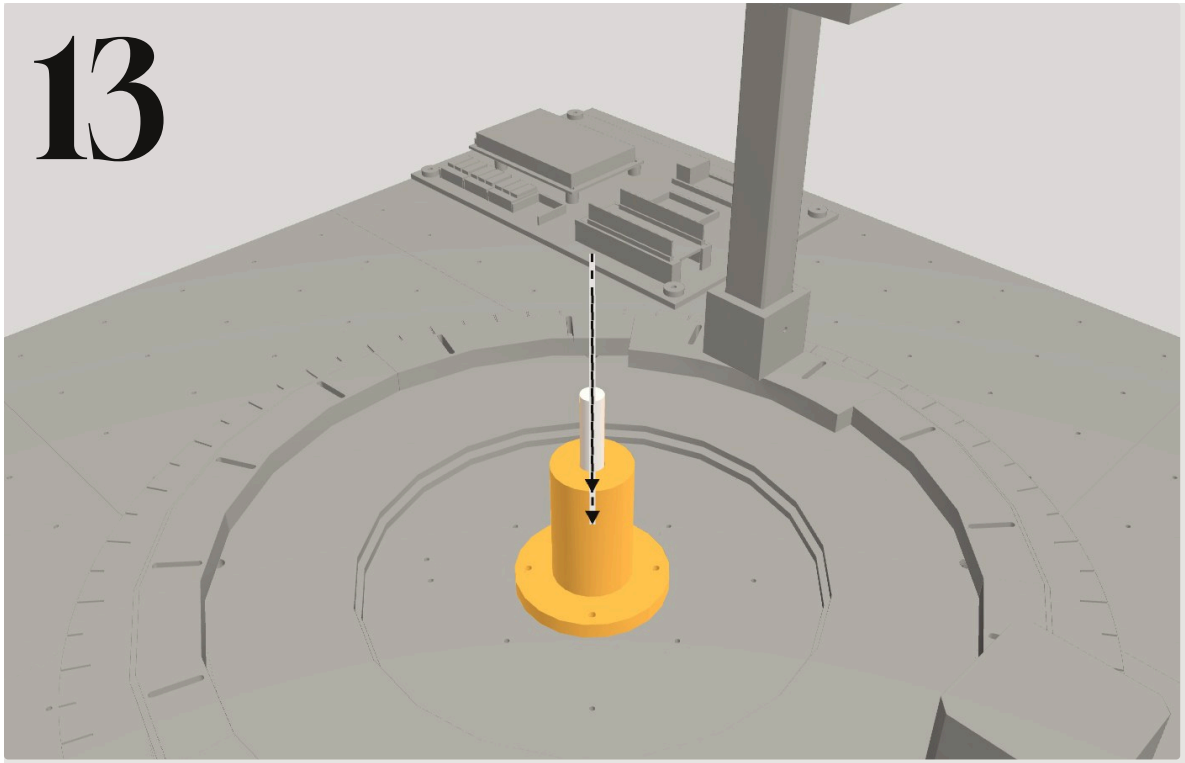
Stage 04 · Light arm (M4) · step 12 of 24

- 1 × Light-arm upright bar, 200 mm LAR-08f · printed · STL
- 1 × Cross clamp: upright to horizontal arm LAR-08e · printed · STL
- 1 × Horizontal light-arm bar, 200 mm LAR-08g · printed · STL

1. The upright stands in the arm segment's socket, 36 mm beside the lamp's beam line (so it never shades the sample).
2. Upright bar in the socket, M2 × 16 through the socket wall into its lower pilot. Cross clamp down the upright until its screw meets the upper pilot (M2 × 16). Arm bar through the clamp's side channel, parallel to the beam line, pointing outward, its inner pilot under the clamp's top screw (M2 × 16). The lamp's clamp takes the outer pilot (M18.1), so the beam centre passes through the zero point, 300 mm out, at 150 mm.

Checklist step M13.5 · Arm (stage 04 · Light arm (M4))

13





1× TBL-07d



1× TBL-05



4× M2 × 12

Table post and axle bolt

Stage 05 · Table (M3) · step 13 of 24

Before touching a motor plug or a screw terminal: press the e-stop and unplug the adapter.

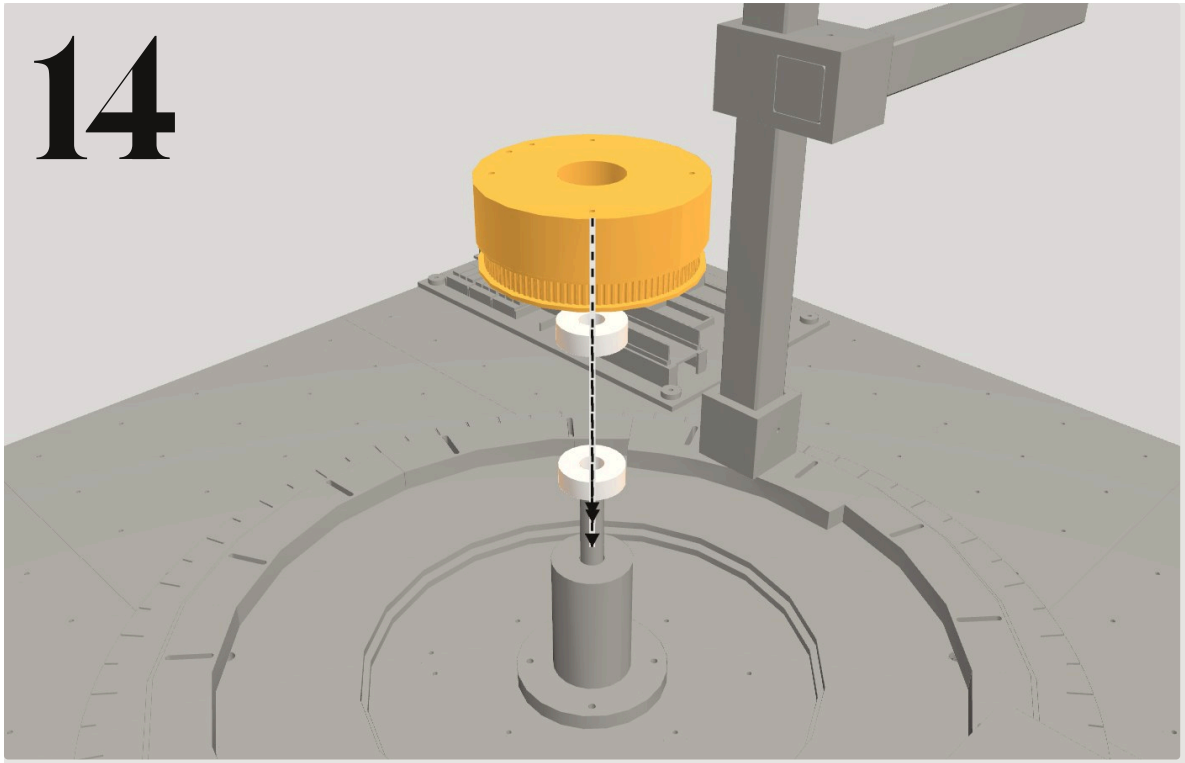
1 × Table post: fixed M8 x 75 axle (hex-locked head) TBL-07d · printed · STL

1 × M8 x 75 hex bolt, partial thread + M8 nut (fixed table axle) TBL-05 · bought

1. Bolt up through the post from below, head into the hex pocket. Post on its 4 pilots at the deck centre; 4 × M2 × 12 (the bolt is now trapped and can't turn).

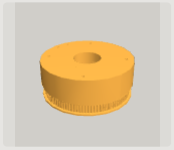
Checklist step M15.2 · Fixed axle and drum (stage 05 · Table (M3))

14





2× FRM-08



1× TBL-07a

Drum on its bearings

Stage 05 · Table (M3) · step 14 of 24

- **2 ×** 608-2RS bearings FRM-08 · bought
- **1 ×** Table drum: 100T teeth, 608 seats, OG-40 top TBL-07a · printed · STL

1. Press both 608s into the drum. Onto the bolt: 1 mm shim (sits on the post top), the drum (lower 608 onto the bolt), 6 mm spacer down the drum's top hole, M8 nut.
2. Tighten the nut with the spanner until the drum has no up-down play but still turns freely (the nut presses only the inner races).

Checklist step M15.2 · Fixed axle and drum (stage 05 · Table (M3))

15

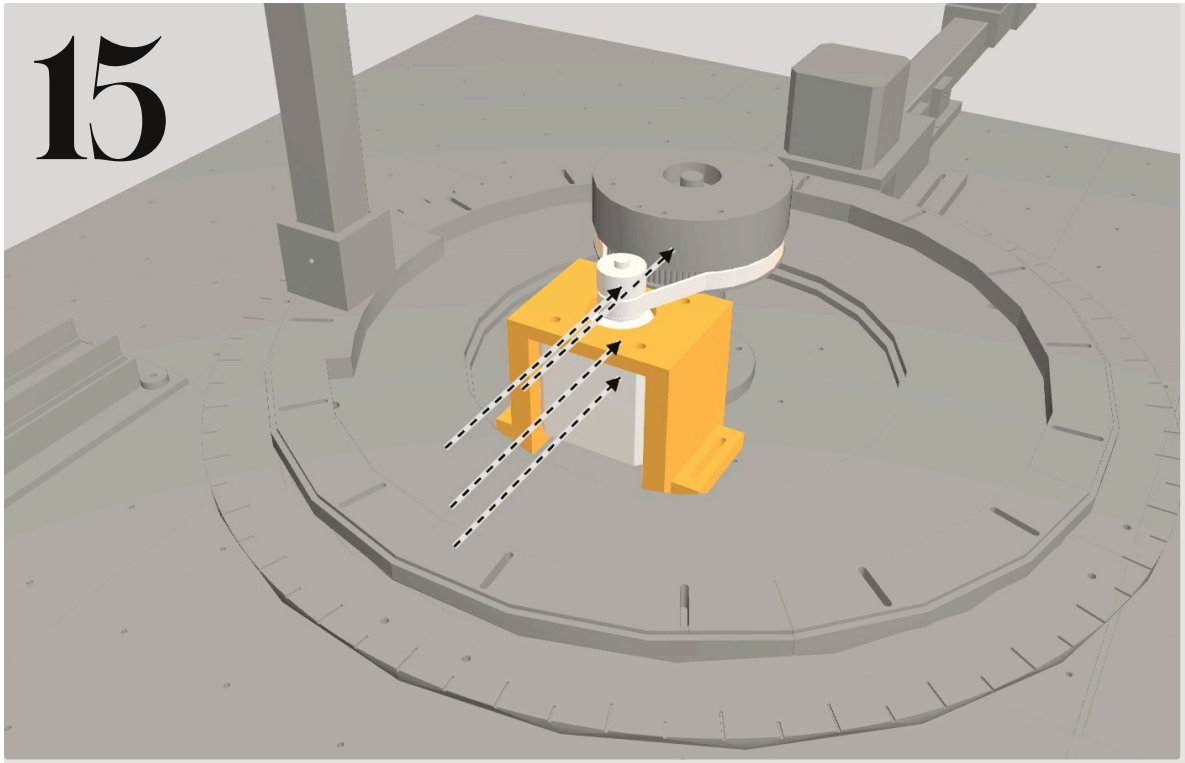


Table motor and belt

Stage 05 · Table (M3) · step 15 of 24

- 1 × Table motor bracket (shaft up, slotted) TBL-07c · printed · STL
- 1 × NEMA 17 stepper 42 Ncm 1.5 A, 38 mm TBL-01 · bought
- 1 × GT2 20T pulley 5 mm bore TBL-03 · bought
- 1 × GT2 6 mm closed belt 232 mm TBL-02 · bought

1. Motor face up, bracket top plate on its face (4 × M3 × 8); 20T pulley on the shaft, teeth level with the drum's teeth.
2. E-stop pressed: its cable into the SKR Pico Y socket, run along the deck edge (never plug or unplug a motor with motor power on).
3. Belt on the motor pulley and the drum's teeth; bracket rails' slots over their 2 deck pilots (inside the lazy Susan, at 180°); pull it outward until the belt deflects ~3 mm under a finger; M2 × 12 + washer through each slot.

Checklist step M15.3 · Drive belt (stage 05 · Table (M3))



1× TBL-07c



1× TBL-01



1× TBL-03



1× TBL-02



M2 × 12



4× M3 × 8

16

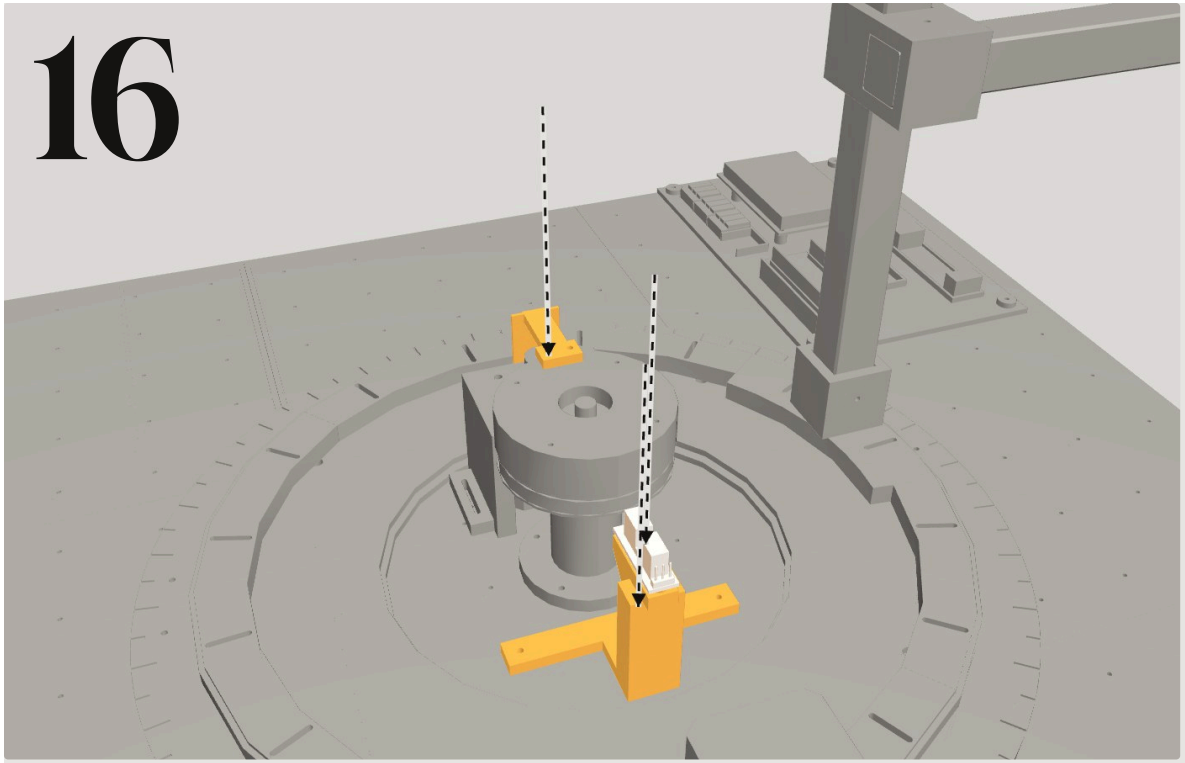


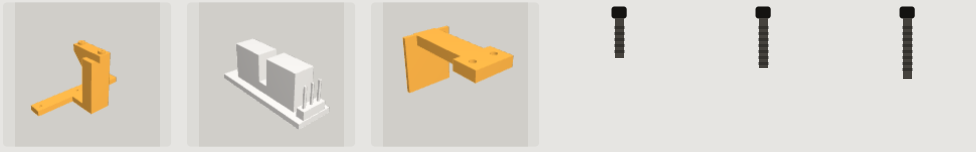
Table home switch

Stage 05 · Table (M3) · step 16 of 24

- **1 ×** Table endstop riser (3 deck pilots, +X) TBL-07g · printed · STL
- **1 ×** Optical endstop module, slot type TBL-04 · bought
- **1 ×** Table home flag (fin at r 60) TBL-07h · printed · STL

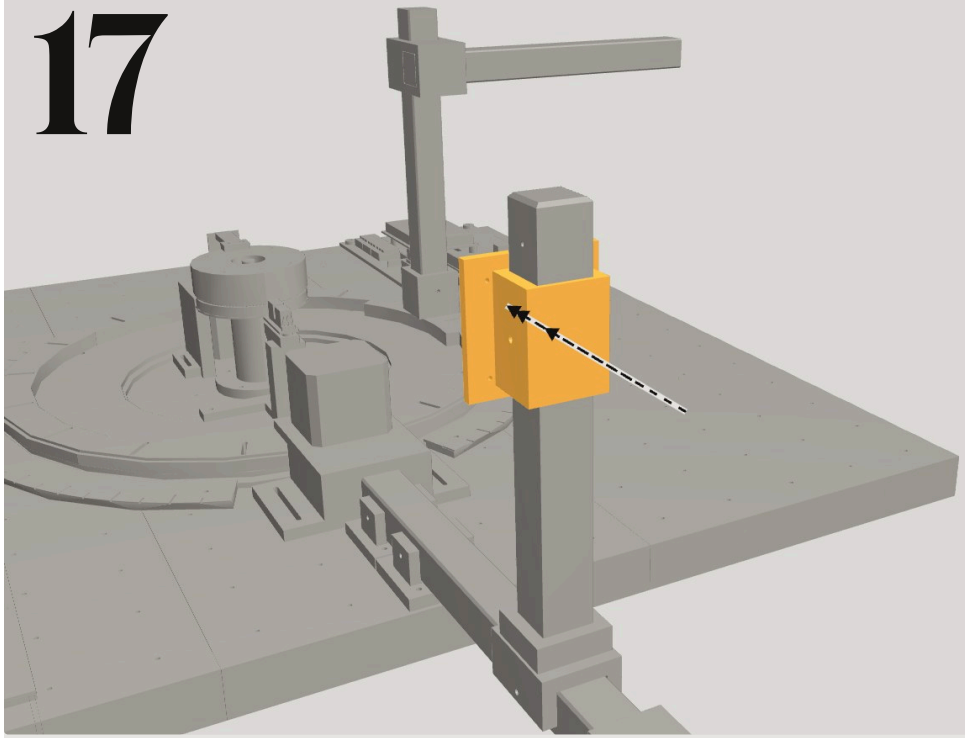
1. Riser on the three deck pilots at x = 40 (y = −40, 0, +40; +X side, inside the lazy Susan opening), 3 × M2 × 10 + washer, post outward. Endstop on its pad: slot over the pad's middle, header end outward, 2 × M2 × 6 self-tap.
2. Flag on the drum top's two flag holes (the close pair outside the four yoke pilots), 2 × M2 × 8 self-tap, fin hanging down outside the drum. Turn the table by hand: at −180° (fin on +X) the fin passes through the slot without touching either side.

Checklist step M16.1 · Flag and homing (stage 05 · Table (M3))



1× TBL-07g **1×** TBL-04 **1×** TBL-07h **2×** M2 × 6 **2×** M2 × 8 **3×** M2 × 10

17





1× CAM-03b



1× CAM-01



1× CAM-02



4× M2 × 8

M2 × 16

Measuring camera

Stage 06 · Camera and light · step 17 of 24

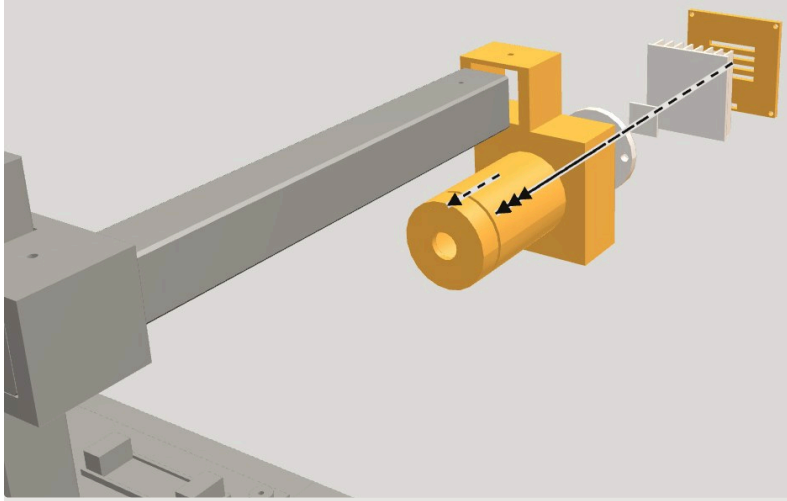
Don't stare into the lamp. Before touching a motor plug or a screw terminal: press the e-stop and unplug the adapter.

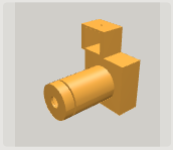
- 1 × Camera plate + post clamp CAM-03b · printed · STL
- 1 × USB camera 12 MP IMX477, M12 mount (UVC) CAM-01 · bought
- 1 × M12 lens 16 mm, ≥ 1/2 in format, low distortion CAM-02 · bought

1. Camera board on the plate: 4 × M2 × 8 (FRM-06) self-tap into the 6 mm standoffs.
2. Plate on the camera pillar: slide its clamp down to the pilot hole, M2 × 16 self-taps into it; aperture stop in front of the lens.

Checklist step M17.2 · Mount (stage 06 · Camera and light)

18





1× LSR-05a



1× LSR-06



1× LSR-01



1× LSR-03



1× LSR-05b



4× M2 × 12



M2 × 16

Lamp

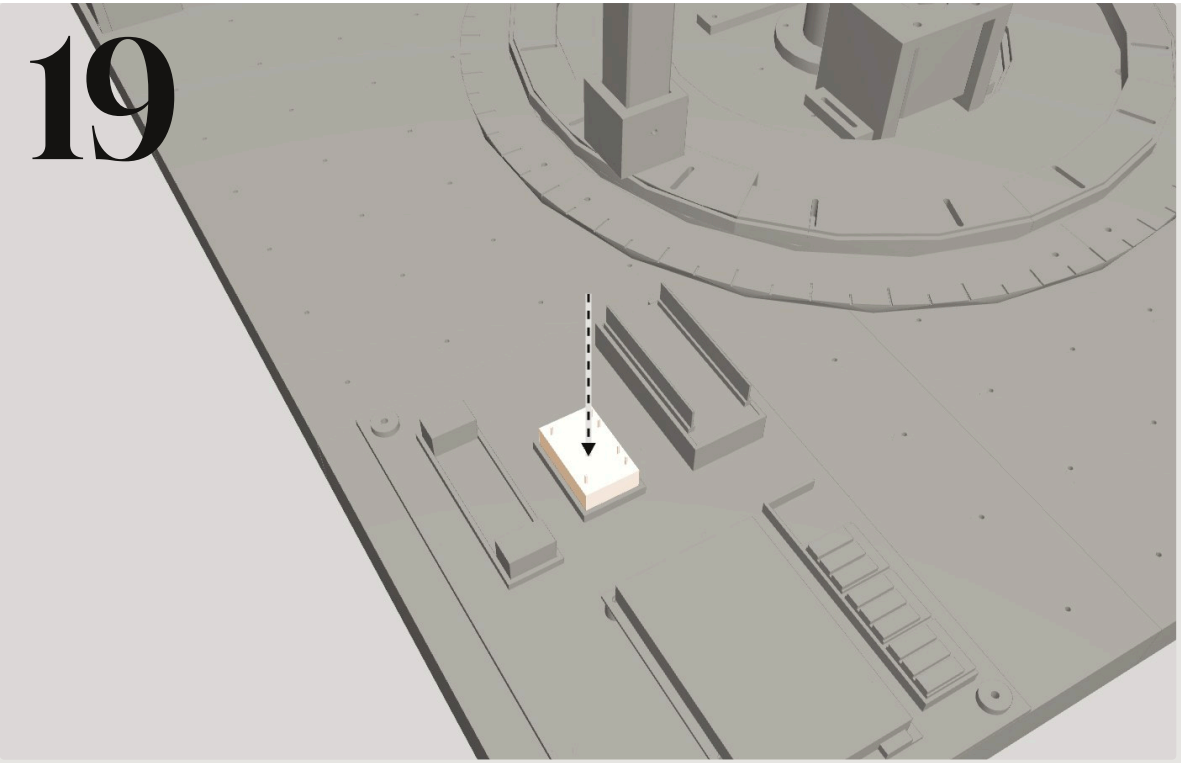
Stage 06 · Camera and light · step 18 of 24

- 1 × Lamp body + tube, bar clamp, polarizer slot LSR-05a · printed · STL
- 1 × Solderless COB holder 13.5 mm (TE LUMAWISE Z35) LSR-06 · bought
- 1 × COB LED CRI 97, 12 V, 13.5 mm (Citizen CLU7A3) LSR-01 · bought
- 1 × Heatsink 40 x 40 mm LSR-03 · bought
- 1 × Lamp back cap LSR-05b · printed · STL

1. Holder face-down in the body's round seat, wire entries toward the lead groove (the side away from the clamp), leads along the groove. COB into the holder's square seat, light-emitting face down through the window. Thermal tape (LSR-04) on the COB's back; heatsink on top, fins out. Cap on with 4 × M2 × 12, leads through its slot; tighten evenly until the heatsink no longer turns.
2. Lamp clamp onto the arm bar's outer pilot (M2 × 16), aperture facing the zero point.

Checklist step M18.1 · Lamp assembly (stage 06 · Camera and light)

19





1× LSR-02

LED driver

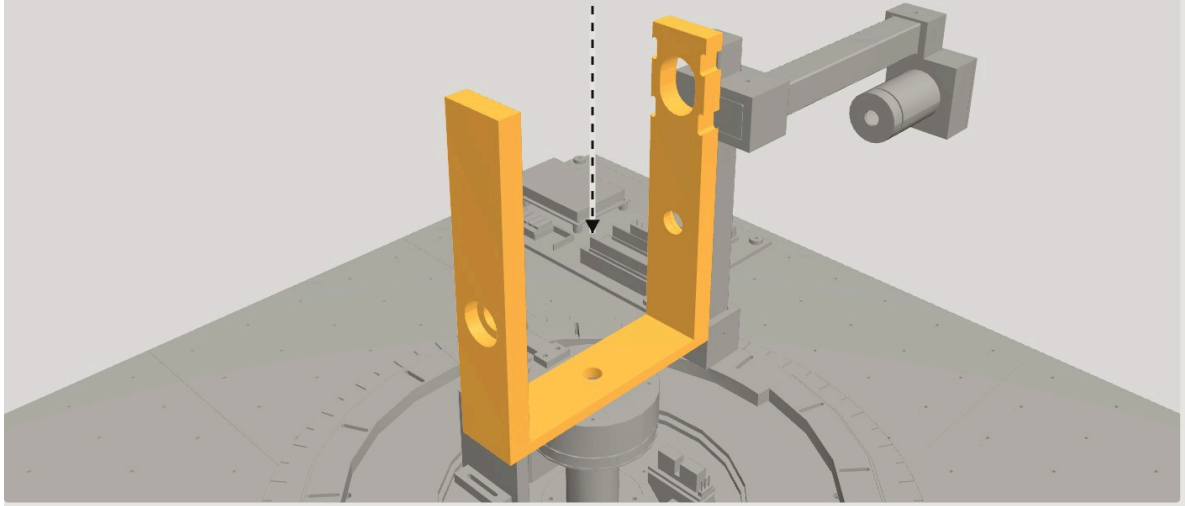
Stage 06 · Camera and light · step 19 of 24

1 × Constant-current driver 350 mA LSR-02 · bought

1. E-stop pressed, adapter unplugged (HE `+` carries motor-power voltage). Driver (LSR-02) pins up in the short frame on the controller tray (double-sided tape). Female jumper ends onto its pins: +Vin ← SKR Pico HE `+` (terminal marked 12/24V; cut jumper, bare end under the screw); −Vin → SKR Pico HE `−` (terminal marked IO23; output port from M08.2, normally `M62 P4`).

Checklist step M18.1 · Lamp assembly (stage 06 · Camera and light)

20





1× TLT-07a



4× M2 × 12

Tilt yoke

Stage 08 · Tilt (M2) · step 20 of 24

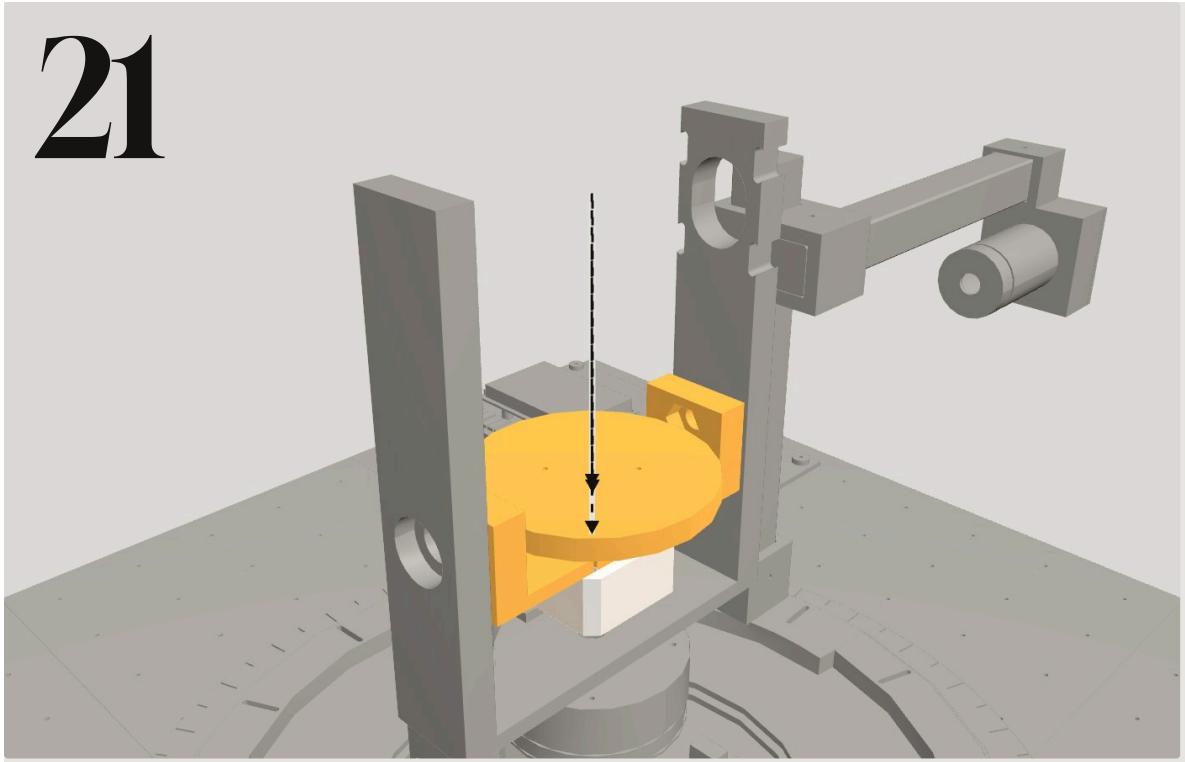
Before touching a motor plug or a screw terminal: press the e-stop and unplug the adapter.

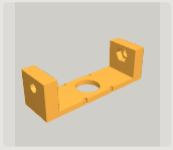
1 × Tilt yoke: 608 seats, motor slots above the axis TLT-07a · printed · STL

1. Yoke on the drum (4 × M2 × 12 self-tap into the OG-40 pilots) with its arms along ±Y: at M3 = 0 the tilt axis is square to the camera direction (+X) and the flag holes on the drum stay clear. The pulley-side arm is +Y.

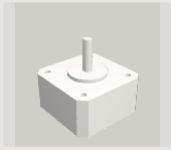
Checklist step M25.2 · Cradle, platter and stub axles (stage 08 · Tilt (M2))

21





1× TLT-07c



1× SPN-01



1× SPN-07a



M2 × 6



4× M3 × 8

Cradle, spin motor and platter

Stage 08 · Tilt (M2) · step 21 of 24

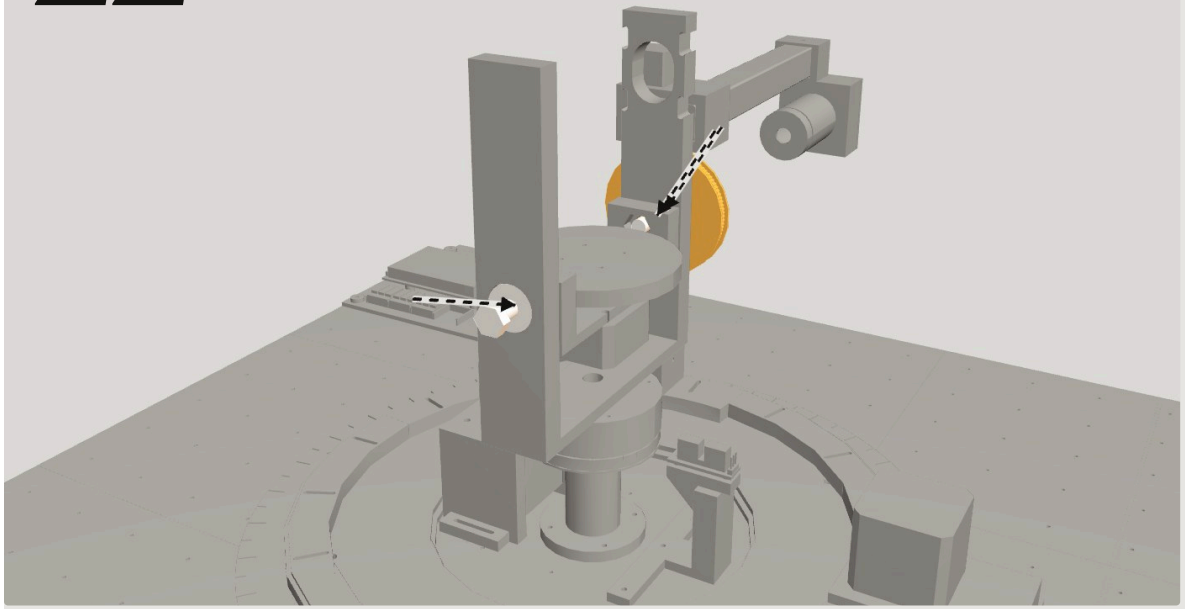
- **1 ×** Tilt cradle: stub-axle nut traps, spin motor floor TLT-07c · printed · STL
- **1 ×** NEMA 17 pancake stepper 16 Ncm 1 A (direct drive) SPN-01 · bought
- **1 ×** Spin platter: clamps the motor D-shaft (direct drive) SPN-07a · printed · STL

1. Pancake motor under the cradle floor (shaft up through it, 4 × M3 × 8). Platter on the shaft: M2 nut in its slot, M2 × 6 through the hub onto the D flat; platter top level with the stub-bolt axis (the end blocks stand 10 mm higher).

Put the motor and platter on the cradle first, then lower the cradle between the yoke arms.

Checklist step M25.2 · Cradle, platter and stub axles (stage 08 · Tilt (M2))

22





1× TLT-07b



2× TLT-05



2× FRM-08

Stub axles and bearings

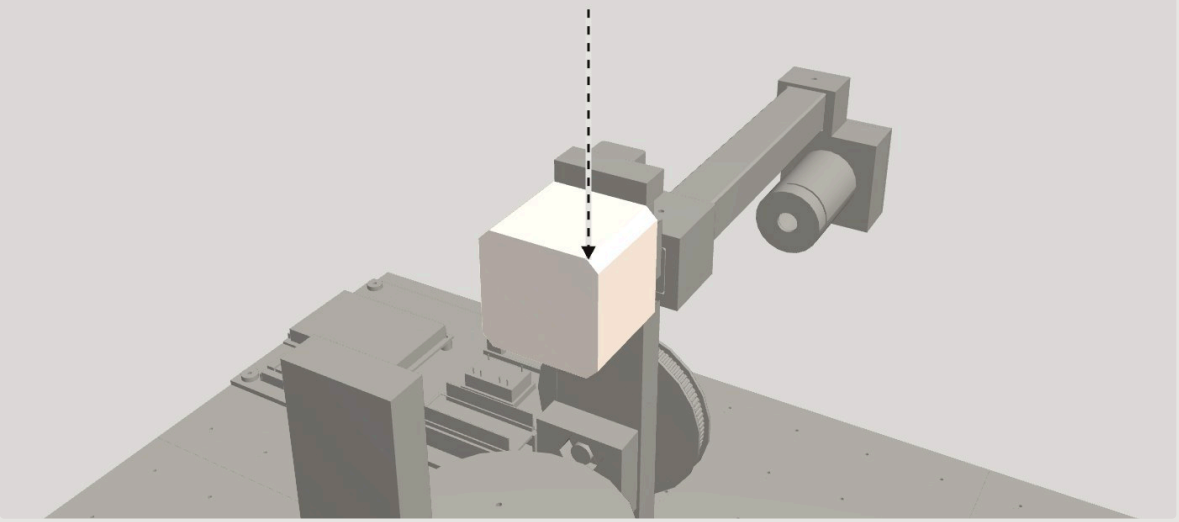
Stage 08 · Tilt (M2) · step 22 of 24

- **1 ×** GT2 100T pulley (5:1) on the stub axle TLT-07b · printed · STL
- **2 ×** M8 x 35 hex bolts, half thread TLT-05 · bought
- **2 ×** 608-2RS bearings FRM-08 · bought

1. Pulley side: bolt through 100T pulley → 1 mm shim → 608 (outer face of the arm) → 6.8 mm spacer → cradle nut. Other side: the same with the 9.4 mm spacer instead of the pulley. Hold each bolt head with the spanner; snug, not tight.

Checklist step M25.2 · Cradle, platter and stub axles (stage 08 · Tilt (M2))

23





1× TLT-01



4× M3 × 16

Tilt motor

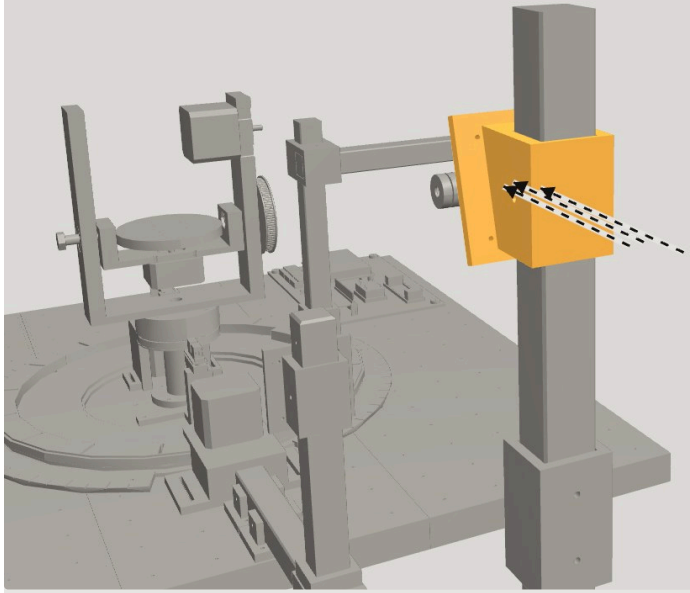
Stage 08 · Tilt (M2) · step 23 of 24

1 × NEMA 17 stepper 42 Ncm 1.5 A, 38 mm TLT-01 · bought

1. E-stop pressed: motor cable into the SKR Pico Z1 socket first, along the yoke arm with slack for the table's ±180°.
2. Motor on the inner face of the pulley-side arm above the axis, shaft out through the arm slot (4 × M3 × 16 in the slots; the arm is 12 mm thick); 20T pulley outside, level with the 100T; belt on; slide the motor up to tension (~3 mm deflection); tighten.

Checklist step M25.3 · Tilt drive and flag (stage 08 · Tilt (M2))

24





1× SCN-03d



1× CAM-04



1× CAM-05



4× M2 × 8

M2 × 16

Scan camera

Stage 11 · 3D scan: capture · step 24 of 24

- **1 ×** Scan camera plate, looks 15 deg down SCN-03d · printed · STL
- **1 ×** Second USB camera 12 MP IMX477, M12 mount (scan camera) CAM-04 · bought
- **1 ×** Second M12 16 mm lens (scan camera) CAM-05 · bought

1. Camera board on the scan plate's standoffs (4 × M2 × 8); plate on the post, M2 × 16 into its pilot: lens centre 168 mm above the zero point (318 mm above the deck), looking 15° down.

Checklist step M31.2 · Mount the scan camera (stage 11 · 3D scan: capture)